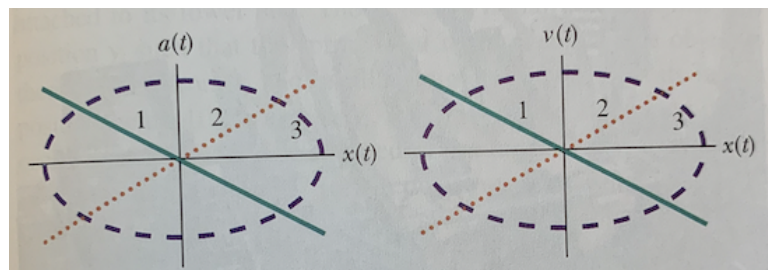


## PYL1002: Waves and Oscillations – Quiz 1 Instructions

1. This quiz is timed and will last **45 min** from 2:05 - 2:50pm.
2. Strict adherence to the **Honor Code** policy of the class is expected. Among other things, you are not allowed to collaborate with others in any form or fashion.
3. There are a total of 2 questions (each with multiple parts). Not all questions are of equal length, so please go through them all before you start and allocate your time appropriately. In particular, problem 1 is a collection of 3 independent, quick conceptual questions which should not require too much math. Some problems may seem lengthy but often the text provided is there to help you.
4. Please justify your answers and show all work.

### 1 Some Quick Conceptual Questions [15 points]

- a [5 points] Consider a particle executing simple harmonic motion (with no dissipation) in one dimension. (i.) Which curve, 1, 2, or 3 in the plot on the left gives the acceleration  $a(t)$  versus position  $x(t)$  of the oscillator? (ii.) Which curve, 1, 2, or 3 in the plot on the right gives the velocity  $v(t)$  versus position  $x(t)$  of the oscillator? Justify your answer in each case.



- b [5 points] Consider the following two roller coaster tracks:

$$\text{Track 1 : } x^2 + (y - 1)^2 = 1, \quad (1)$$

$$\text{Track 2 : } x^2 + 4(y - 1)^2 = 4. \quad (2)$$

If I am looking at small oscillations around the lowest point on each of the tracks, which is at  $(0, 0)$ , in which case will I have a higher oscillation frequency and why? Justify your answer using the physics of small oscillations we have learned in this class. You can ignore damping.

- c [5 points] A child of mass  $m = 30$  kg sits on a playground swing of length  $l = 2.5$  m. An adult gives small periodic pushes to make the swing go higher and higher. *Roughly* how many times per minute should the adult push the swing for maximum effectiveness? You can assume small oscillations and neglect air resistance and other damping effects. You may take  $g \approx 10 \text{ m/s}^2$  and  $\pi \approx 3$ .

## 2 NASA Wants Your Help! [10 points]

In orbit around the Earth, an astronaut is practically “weightless” due to being in freefall. Being weightless means you float in the space shuttle<sup>1</sup> which is all fun, but it also means you can’t just stand on a weighing scale to measure your weight (and correspondingly mass). How do astronauts measure their mass in space then? Help NASA figure this out.

You are designing a *body-mass measuring device* (BMMD) which is designed for use on orbiting space vehicles. The BMMD is a spring-mounted chair, on which an astronaut measures his or her period of oscillation, and the mass then follows from the physics behind oscillating spring systems. See figure below of NASA astronaut Alan Bean on the BMMD to get a sense.

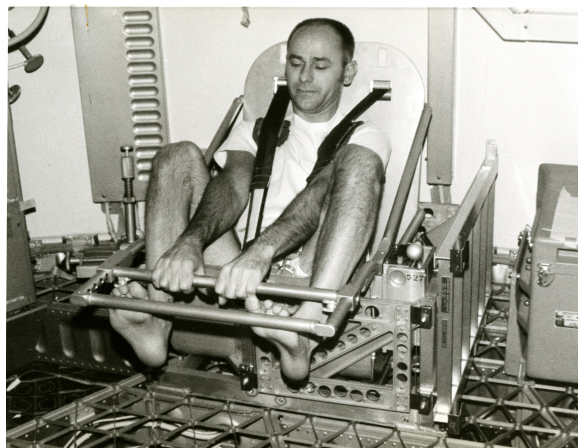


Figure 1: Astronaut Alan Bean in a Body Mass Measurement Device (BMMD). Image courtesy: UTMB Moody Medical Library

- a [2 points] If  $M$  is the mass of the astronaut, and  $m$  is the effective mass of the chair and other contraptions on the oscillating part of the BMMD, show that if one measures the time period  $T$  of oscillation and knows the spring constant  $k$ , then we can estimate the mass of the astronaut as follows,

$$M = \left( \frac{k}{4\pi^2} \right) T^2 - m, \quad (3)$$

- b [1 point] Let us now look at the BMMD on NASA’s Skylab Mission Two. The spring constant on this machine was about  $k = 40\pi^2$  N/m, and period of the oscillation of the *empty* chair was 1 sec. Calculate the effective mass  $m$  of the chair.
- c [1 point] With astronaut Alan Bean in the chair, the period of oscillation became 3 sec. Calculate the mass of Mr. Bean.
- d [3 points] Due to repeated use, the BMMD has worn off a bit which has introduced a damping force  $F_{\text{damp}} = -b\dot{x}$  where  $b$  is a positive constant. To ensure the BMMD still allows for sustained oscillations (that is, it is *underdamped*), what is the minimum mass of the astronaut that can be measured (in terms of other variables such as  $k, m, b$  etc)?
- e [3 points] For  $b = 60\pi$  kg/s, would the time period of oscillation change if Astronaut Bean repeated the measurement of part (c)? If yes, give an estimate of the new time period? If needed *at the end*, you may use  $\sqrt{3} \approx 1.7$  and  $\pi \approx 3$ .

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<sup>1</sup>which does look like a lot of fun, for instance on this YouTube video (<https://youtu.be/kVK20xyfPrU?si=V2XMqYBgP2ZZpwkd>) which you can see after the quiz, featuring a fly-through of the International Space Shuttle